

474_Project 1

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Introduction

The goal of this project was to compare the predictive performance of several classification models on real-world financial data from UCI's repo specifically to predict company bankruptcies. The models evaluated include: GLM, KNN, and Naive Bayes, and their performance was assessed using 5-fold cross-validation. The focus was on both overall accuracy and the ability to correctly identify bankrupt companies (recall).

Methods

1 Data Preprocessing

The data was cleaned and standardized with numeric columns, near-zero variance predictors were removed, and categorical variables were properly formatted.

2 Modeling

Fitted GLM (logistic regression), KNN, and Naive Bayes, then used a 5-fold cross validation to estimate out-of-sample performance.

3 Evaluation Metrics

Accuracy: overall proportion of correct predictors.

Precision: proportion of predicted bankruptcies that were correct.

Recall: proportion of actual bankruptcies correctly predicted.

All modeling code is included in the appendix.

Results

Summary: GLM and KNN had the highest accuracy but very low recall for bankruptcies while Naive Bayes had slightly lower overall accuracy but much higher recall, indicating it is better at identifying bankruptcies.

Conclusion

In conclusion, the best overall model depends on the goal. If correct predictions was critical then Naive Bayes would be preferred, if overall accuracy is critical then GLM or KNN would suffice.

The practical takeaways from this project are that cross-validation helped reveal the tradeoffs between accuracy and recall, and that there is a balance between interpretability and predictor performance.

Appendix

Data Preprocessing

```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.4.1
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
## Warning: package 'tibble' was built under R version 4.4.1
```

```
## Warning: package 'tidyr' was built under R version 4.4.1
```

```
## Warning: package 'readr' was built under R version 4.4.1
```

```
## Warning: package 'purrr' was built under R version 4.4.1
```

```
## Warning: package 'dplyr' was built under R version 4.4.1
```

```
## Warning: package 'stringr' was built under R version 4.4.1
```

```
## Warning: package 'forcats' was built under R version 4.4.1
```

```
## Warning: package 'lubridate' was built under R version 4.4.3
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats   1.0.0      v stringr   1.5.1
```

```
## v ggplot2   3.5.2      v tibble    3.2.1
```

```
## v lubridate 1.9.4      v tidyr     1.3.1
```

```
## v purrr     1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(caret)
```

```
## Warning: package 'caret' was built under R version 4.4.3
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```
library(e1071) #for naivebayes
```

```
## Warning: package 'e1071' was built under R version 4.4.3
```

```
library(MASS) #for lda, qda
```

```
## Warning: package 'MASS' was built under R version 4.4.3
```

```
##
## Attaching package: 'MASS'
##
## The following object is masked from 'package:dplyr':
##
##     select
```

```
library(class) #for knn
```

```
## Warning: package 'class' was built under R version 4.4.3
```

```
data <- read.csv("data.csv", fileEncoding = "latin1")
```

```
names(data) <- gsub("[^A-Za-z0-9_]", ".", names(data))
names(data) <- make.names(names(data), unique=T)
```

```
for(col in names(data)){
  if(is.character(data[[col]])){
    data[[col]] <- gsub("Â%", "", data[[col]], fixed=T)
    data[[col]] <- gsub("%", "", data[[col]], fixed=T)
    data[[col]] <- gsub("\\\\?", "", data[[col]], fixed=T)
  }
}
```

```
data$Bankrupt. <- as.factor(data$Bankrupt.)
```

```
num_cols <- sapply(data, is.numeric)
data[num_cols] <- as.data.frame(scale(data[num_cols]))
```

```
data[num_cols] <- lapply(data[num_cols], function(x){
  q_low <- quantile(x,0.01,na.rm = T)
  q_high <- quantile(x,0.99, na.rm = T)
  x[x < q_low] <-q_low
  x[x > q_high] <- q_high
  return(x)
})
```

```
data <- data[,sapply(data,function(x) length(unique(x)) >1)]
```

Data Exploration

```
dim(data)
```

```
## [1] 6819 94
```

```
table(data$Bankrupt.)
```

```
##
##      0      1
## 6599 220
```

```
prop.table(table(data$Bankrupt.))
```

```
##
##           0           1
## 0.9677372 0.0322628
```

```
summary(data)
```

```
## Bankrupt. ROA.C..before.interest.and.depreciation.before.interest
## 0:6599 Min. :-2.756882
## 1: 220 1st Qu.: -0.472147
##      Median :-0.040768
##      Mean  : 0.004824
##      3rd Qu.: 0.500665
##      Max.   : 2.623388
## ROA.A..before.interest.and...after.tax
## Min.    :-3.141505
## 1st Qu. :-0.351752
## Median : 0.017931
## Mean    : 0.007665
## 3rd Qu. : 0.465289
## Max.    : 2.469320
## ROA.B..before.interest.and.depreciation.after.tax Operating.Gross.Margin
## Min.    :-2.950853 Min.    :-1.60355
## 1st Qu. :-0.427180 1st Qu. :-0.44310
## Median :-0.021280 Median :-0.11519
```

## Mean : 0.006613	Mean : 0.01682
## 3rd Qu.: 0.495438	3rd Qu.: 0.35232
## Max. : 2.513292	Max. : 2.60245
## Realized.Sales.Gross.Margin	Operating.Profit.Rate
## Min. : -1.57665	Min. : -0.09017
## 1st Qu.: -0.44311	1st Qu.: 0.01645
## Median : -0.11549	Median : 0.02053
## Mean : 0.01709	Mean : 0.01923
## 3rd Qu.: 0.34953	3rd Qu.: 0.02609
## Max. : 2.60466	Max. : 0.04981
## After.tax.net.Interest.Rate	Pre.tax.net.Interest.Rate
## Min. : -0.13002	Min. : -0.15324
## 1st Qu.: 0.01676	1st Qu.: 0.01524
## Median : 0.02144	Median : 0.02128
## Mean : 0.01866	Mean : 0.01840
## 3rd Qu.: 0.02836	3rd Qu.: 0.03024
## Max. : 0.06493	Max. : 0.07513
## Non.industry.income.and.expenditure.revenue	
## Min. : -0.127671	
## 1st Qu.: -0.014032	
## Median : -0.008728	
## Mean : -0.008955	
## 3rd Qu.: -0.003380	
## Max. : 0.107251	
## Continuous.interest.rate..after.tax.	Operating.Expense.Rate
## Min. : -0.14303	Min. : -0.6162885
## 1st Qu.: 0.01462	1st Qu.: -0.6162885
## Median : 0.02000	Median : -0.6162885
## Mean : 0.01709	Mean : -0.0003152
## 3rd Qu.: 0.02792	3rd Qu.: 0.6639477
## Max. : 0.06992	Max. : 2.4074780
## Research.and.development.expense.rate	Cash.flow.rate
## Min. : -0.750658	Min. : -1.753983
## 1st Qu.: -0.750658	1st Qu.: -0.344776
## Median : -0.554760	Median : -0.138033
## Mean : -0.001155	Mean : -0.008164
## 3rd Qu.: 0.577138	3rd Qu.: 0.209723
## Max. : 2.882499	Max. : 2.886489
## Interest.bearing.debt.interest.rate	Tax.rate..A.
## Min. : -0.15191	Min. : -0.82933
## 1st Qu.: -0.15191	1st Qu.: -0.82933
## Median : -0.15191	Median : -0.29936
## Mean : -0.01132	Mean : -0.01229
## 3rd Qu.: -0.15191	3rd Qu.: 0.65509
## Max. : 6.85063	Max. : 3.81171
## Net.Value.Per.Share..A.	Net.Value.Per.Share..B.
## Min. : -1.51180	Min. : -1.50185
## 1st Qu.: -0.50848	1st Qu.: -0.51058
## Median : -0.18621	Median : -0.18749
## Mean : -0.01563	Mean : -0.01623
## 3rd Qu.: 0.26699	3rd Qu.: 0.26684
## Max. : 3.57768	Max. : 3.58583
## Net.Value.Per.Share..A.	Net.Value.Per.Share..C.
## Min. : -1.51267	
## 1st Qu.: -0.50766	
## Median : -0.18734	
## Mean : -0.01562	
## 3rd Qu.: 0.26702	
## Max. : 3.57580	
## Persistent.EPS.in.the.Last.Four.Seasons	Cash.Flow.Per.Share
## Min. : -1.95269	Min. : -2.356928
## 1st Qu.: -0.42395	1st Qu.: -0.325603
## Median : -0.12834	Median : -0.056489
## Mean : -0.01533	Mean : -0.005297
## 3rd Qu.: 0.30086	3rd Qu.: 0.291953
## Max. : 3.10582	Max. : 2.528608
## Revenue.Per.Share..Yuan....	Operating.Profit.Per.Share..Yuan....
## Min. : -0.0257	Min. : -1.51937

## 1st Qu.:-0.0257	1st Qu.:-0.46551	
## Median :-0.0257	Median :-0.17410	
## Mean :-0.0257	Mean :-0.01664	
## 3rd Qu.:-0.0257	3rd Qu.: 0.25282	
## Max. :-0.0257	Max. : 3.27515	
## Per.Share.Net.profit.before.tax..Yuan....		
## Min. :-1.8561		
## 1st Qu.:-0.4217		
## Median :-0.1402		
## Mean :-0.0184		
## 3rd Qu.: 0.2752		
## Max. : 2.9656		
## Realized.Sales.Gross.Profit.Growth.Rate Operating.Profit.Growth.Rate		
## Min. :-0.06098	Min. :-0.1482635	
## 1st Qu.:-0.02842	1st Qu.: 0.0003825	
## Median :-0.02529	Median : 0.0059092	
## Mean :-0.02036	Mean : 0.0101708	
## 3rd Qu.:-0.02109	3rd Qu.: 0.0132570	
## Max. : 0.17204	Max. : 0.2722136	
## After.tax.Net.Profit.Growth.Rate Regular.Net.Profit.Growth.Rate		
## Min. :-0.768983	Min. :-0.764982	
## 1st Qu.: 0.008938	1st Qu.: 0.008645	
## Median : 0.021108	Median : 0.020743	
## Mean : 0.016844	Mean : 0.016327	
## 3rd Qu.: 0.036169	3rd Qu.: 0.035729	
## Max. : 0.587399	Max. : 0.548774	
## Continuous.Net.Profit.Growth.Rate Total.Asset.Growth.Rate		
## Min. :-0.102082	Min. :-1.9008396	
## 1st Qu.:-0.005913	1st Qu.:-0.2236576	
## Median :-0.004071	Median : 0.3077951	
## Mean :-0.004184	Mean :-0.0004081	
## 3rd Qu.:-0.001740	3rd Qu.: 0.6494433	
## Max. : 0.091455	Max. : 1.4667072	
## Net.Value.Growth.Rate Total.Asset.Return.Growth.Rate.Ratio Cash.Reinvestment..		
## Min. :-0.01372	Min. :-0.24613	Min. :-2.823319
## 1st Qu.:-0.01372	1st Qu.:-0.05071	1st Qu.:-0.237655
## Median :-0.01372	Median :-0.02055	Median : 0.036110
## Mean :-0.01372	Mean :-0.01178	Mean : 0.003224
## 3rd Qu.:-0.01372	3rd Qu.: 0.01462	3rd Qu.: 0.340194
## Max. :-0.01372	Max. : 0.36434	Max. : 1.993403
## Current.Ratio Quick.Ratio Interest.Expense.Ratio		
## Min. :-0.01211	Min. :-0.03423	Min. :-0.601018
## 1st Qu.:-0.01211	1st Qu.:-0.03423	1st Qu.:-0.033702
## Median :-0.01211	Median :-0.03423	Median :-0.026054
## Mean :-0.01211	Mean :-0.03423	Mean :-0.003684
## 3rd Qu.:-0.01211	3rd Qu.:-0.03423	3rd Qu.: 0.011945
## Max. :-0.01211	Max. :-0.03423	Max. : 0.649387
## Total.debt.Total.net.worth Debt.ratio.. Net.worth.Assets		
## Min. :-0.02622	Min. :-1.836567	Min. :-2.327329
## 1st Qu.:-0.02622	1st Qu.:-0.747150	1st Qu.:-0.660739
## Median :-0.02622	Median :-0.032833	Median : 0.032833
## Mean :-0.02622	Mean :-0.006383	Mean : 0.006383
## 3rd Qu.:-0.02622	3rd Qu.: 0.660739	3rd Qu.: 0.747150
## Max. :-0.02622	Max. : 2.327329	Max. : 1.836567

```

## Long.term.fund.suitability.ratio..A. Borrowing.dependency
## Min.      :-0.13754          Min.      :-0.30806
## 1st Qu.   :-0.12571          1st Qu.   :-0.27548
## Median    :-0.11076          Median    :-0.12464
## Mean      :-0.04582          Mean      :-0.02345
## 3rd Qu.   :-0.06874          3rd Qu.   : 0.09925
## Max.      : 1.40693          Max.      : 1.56860
## Contingent.liabilities.Net.worth Operating.profit.Paid.in.capital
## Min.      :-0.04943          Min.      :-1.51084
## 1st Qu.   :-0.04943          1st Qu.   :-0.46333
## Median    :-0.04943          Median    :-0.17434
## Mean      :-0.01711          Mean      :-0.01716
## 3rd Qu.   :-0.01673          3rd Qu.   : 0.25019
## Max.      : 0.33428          Max.      : 3.25262
## Net.profit.before.tax.Paid.in.capital
## Min.      :-1.79303
## 1st Qu.   :-0.43329
## Median    :-0.13836
## Mean      :-0.01534
## 3rd Qu.   : 0.28884
## Max.      : 3.06037
## Inventory.and.accounts.receivable.Net.value Total.Asset.Turnover
## Min.      :-0.64077          Min.      :-1.31109
## 1st Qu.   :-0.37951          1st Qu.   :-0.64406
## Median    :-0.17474          Median    :-0.22903
## Mean      :-0.02007          Mean      :-0.01302
## 3rd Qu.   : 0.15697          3rd Qu.   : 0.34906
## Max.      : 2.38223          Max.      : 3.77046
## Accounts.Receivable.Turnover Average.Collection.Days
## Min.      :-0.04596          Min.      :-0.03833
## 1st Qu.   :-0.04596          1st Qu.   :-0.03833
## Median    :-0.04596          Median    :-0.03833
## Mean      :-0.04596          Mean      :-0.03833
## 3rd Qu.   :-0.04596          3rd Qu.   :-0.03833
## Max.      :-0.04596          Max.      :-0.03833
## Inventory.Turnover.Rate..times. Fixed.Assets.Turnover.Frequency
## Min.      :-0.6616773        Min.      :-0.407093
## 1st Qu.   :-0.6616773        1st Qu.   :-0.407093
## Median    :-0.6616773        Median    :-0.407093
## Mean      :-0.0004348        Mean      :-0.001368
## 3rd Qu.   : 0.7607509        3rd Qu.   :-0.407093
## Max.      : 2.3340428        Max.      : 3.411184
## Net.Worth.Turnover.Rate..times. Revenue.per.person Operating.profit.per.person
## Min.      :-0.76999          Min.      :-0.01702   Min.      :-1.32598
## 1st Qu.   :-0.45858          1st Qu.   :-0.01702   1st Qu.   :-0.25162
## Median    :-0.24752          Median    :-0.01702   Median    :-0.14588
## Mean      :-0.02735          Mean      :-0.01702   Mean      :-0.03194
## 3rd Qu.   : 0.11745          3rd Qu.   :-0.01702   3rd Qu.   : 0.03606
## Max.      : 3.73080          Max.      :-0.01702   Max.      : 3.19128
## Allocation.rate.per.person Working.Capital.to.Total.Assets
## Min.      :-0.03822          Min.      :-2.196616
## 1st Qu.   :-0.03822          1st Qu.   :-0.674229
## Median    :-0.03822          Median    :-0.065193
## Mean      :-0.03822          Mean      : 0.004662

```

## 3rd Qu.:-0.03822	3rd Qu.: 0.613971	
## Max. :-0.03822	Max. : 2.465712	
## Quick.Assets.Total.Assets	Current.Assets.Total.Assets	Cash.Total.Assets
## Min. :-1.77878	Min. :-2.0012515	Min. :-0.872146
## 1st Qu.:-0.78297	1st Qu.:-0.7767943	1st Qu.:-0.650276
## Median :-0.06773	Median :-0.0341277	Median :-0.353371
## Mean :-0.00142	Mean : 0.0008461	Mean :-0.006914
## 3rd Qu.: 0.69536	3rd Qu.: 0.7646410	3rd Qu.: 0.265554
## Max. : 2.30876	Max. : 1.9984057	Max. : 3.967322
## Quick.Assets.Current.Liability	Cash.Current.Liability	
## Min. :-0.02094	Min. :-0.07281	
## 1st Qu.:-0.02094	1st Qu.:-0.07281	
## Median :-0.02094	Median :-0.07281	
## Mean :-0.02094	Mean :-0.07281	
## 3rd Qu.:-0.02094	3rd Qu.:-0.07281	
## Max. :-0.02094	Max. :-0.07281	
## Current.Liability.to.Assets	Operating.Funds.to.Liability	
## Min. :-1.570283	Min. :-1.969332	
## 1st Qu.:-0.743122	1st Qu.:-0.364326	
## Median :-0.158442	Median :-0.148841	
## Mean :-0.006059	Mean :-0.007892	
## 3rd Qu.: 0.573678	3rd Qu.: 0.201634	
## Max. : 2.689761	Max. : 3.243259	
## Inventory.Working.Capital	Inventory.Current.Liability	
## Min. :-0.39514	Min. :-0.09588	
## 1st Qu.:-0.03450	1st Qu.:-0.09588	
## Median :-0.02077	Median :-0.09588	
## Mean :-0.00527	Mean :-0.06796	
## 3rd Qu.: 0.00321	3rd Qu.:-0.09588	
## Max. : 0.59790	Max. : 2.33621	
## Current.Liabilities.Liability	Working.Capital.Equity	
## Min. :-2.689894	Min. :-1.08056	
## 1st Qu.:-0.651346	1st Qu.:-0.18879	
## Median : 0.219098	Median : 0.01680	
## Mean : 0.003581	Mean : 0.01193	
## 3rd Qu.: 0.872995	3rd Qu.: 0.23492	
## Max. : 1.153497	Max. : 0.69767	
## Current.Liabilities.Equity	Long.term.Liability.to.Current.Assets	
## Min. :-0.36449	Min. :-0.09497	
## 1st Qu.:-0.24570	1st Qu.:-0.09497	
## Median :-0.12787	Median :-0.09497	
## Mean :-0.02134	Mean :-0.06543	
## 3rd Qu.: 0.06766	3rd Qu.:-0.09497	
## Max. : 1.67875	Max. : 2.35685	
## Retained.Earnings.to.Total.Assets	Total.income.Total.expense	
## Min. :-3.15874	Min. :-0.059020	
## 1st Qu.:-0.14224	1st Qu.:-0.025912	
## Median : 0.11499	Median :-0.017595	
## Mean : 0.02657	Mean :-0.013376	
## 3rd Qu.: 0.39424	3rd Qu.:-0.004721	
## Max. : 1.40098	Max. : 0.096885	
## Total.expense.Assets	Current.Asset.Turnover.Rate	Quick.Asset.Turnover.Rate
## Min. :-0.9168	Min. :-0.4238878	Min. :-0.6411173
## 1st Qu.:-0.5384	1st Qu.:-0.4238878	1st Qu.:-0.6411173

## Median :-0.2398	Median :-0.4238878	Median :-0.6411173
## Mean :-0.0197	Mean :-0.0004062	Mean :-0.0002711
## 3rd Qu.: 0.2485	3rd Qu.: -0.4238878	3rd Qu.: 0.8107585
## Max. : 3.6729	Max. : 3.0456764	Max. : 2.2715233
## Working.capitcal.Turnover.Rate	Cash.Turnover.Rate	Cash.Flow.to.Sales
## Min. :-0.0237357	Min. :-0.8412024	Min. :-0.016706
## 1st Qu.: -0.0080189	1st Qu.: -0.8412024	1st Qu.: 0.003691
## Median :-0.0048551	Median :-0.4736834	Median : 0.004622
## Mean :-0.0027146	Mean :-0.0006448	Mean : 0.004879
## 3rd Qu.: -0.0004375	3rd Qu.: 0.6935299	3rd Qu.: 0.005973
## Max. : 0.0597847	Max. : 2.4324394	Max. : 0.026149
## Fixed.Assets.to.Assets	Current.Liability.to.Liability	
## Min. :-0.01211	Min. :-2.689894	
## 1st Qu.: -0.01211	1st Qu.: -0.651346	
## Median :-0.01211	Median : 0.219098	
## Mean :-0.01211	Mean : 0.003581	
## 3rd Qu.: -0.01211	3rd Qu.: 0.872995	
## Max. :-0.01211	Max. : 1.153497	
## Current.Liability.to.Equity	Equity.to.Long.term.Liability	
## Min. :-0.36449	Min. :-0.24125	
## 1st Qu.: -0.24570	1st Qu.: -0.24125	
## Median :-0.12787	Median :-0.16922	
## Mean :-0.02134	Mean :-0.02979	
## 3rd Qu.: 0.06766	3rd Qu.: 0.07483	
## Max. : 1.67875	Max. : 1.34184	
## Cash.Flow.to.Total.Assets	Cash.Flow.to.Liability	CFO.to.Assets
## Min. :-2.663274	Min. :-2.2656723	Min. :-2.804820
## 1st Qu.: -0.347573	1st Qu.: -0.1580612	1st Qu.: -0.468372
## Median :-0.092124	Median :-0.0701045	Median :-0.002541
## Mean : 0.004022	Mean : 0.0000081	Mean : 0.002759
## 3rd Qu.: 0.281416	3rd Qu.: 0.0797054	3rd Qu.: 0.535408
## Max. : 3.299773	Max. : 3.0163223	Max. : 2.423781
## Cash.Flow.to.Equity	Current.Liability.to.Current.Assets	
## Min. :-1.573957	Min. :-0.89859	
## 1st Qu.: -0.199654	1st Qu.: -0.43679	
## Median :-0.048580	Median :-0.12674	
## Mean :-0.002984	Mean :-0.03605	
## 3rd Qu.: 0.163939	3rd Qu.: 0.22267	
## Max. : 1.746450	Max. : 2.64337	
## Net.Income.to.Total.Assets	Total.assets.to.GNP.price	No.credit.Interval
## Min. :-3.17327	Min. :-0.04949	Min. :-0.690755
## 1st Qu.: -0.27299	1st Qu.: -0.04949	1st Qu.: -0.022643
## Median : 0.07088	Median :-0.04949	Median :-0.002876
## Mean : 0.01906	Mean :-0.04949	Mean :-0.004423
## 3rd Qu.: 0.46351	3rd Qu.: -0.04949	3rd Qu.: 0.020637
## Max. : 1.99657	Max. :-0.04949	Max. : 0.769960
## Gross.Profit.to.Sales	Net.Income.to.Stockholder.s.Equity	Liability.to.Equity
## Min. :-1.60365	Min. :-0.98498	Min. :-0.36388
## 1st Qu.: -0.44310	1st Qu.: -0.01978	1st Qu.: -0.23652
## Median :-0.11504	Median : 0.05348	Median :-0.10977
## Mean : 0.01682	Mean : 0.02720	Mean :-0.02553
## 3rd Qu.: 0.35237	3rd Qu.: 0.13461	3rd Qu.: 0.07495
## Max. : 2.60237	Max. : 0.41474	Max. : 1.36438
## Degree.of.Financial.Leverage..DFL.		

```
## Min.      :-0.14161
## 1st Qu.   :-0.04787
## Median    :-0.04678
## Mean      :-0.03184
## 3rd Qu.   :-0.04008
## Max.      : 0.57219
## Interest.Coverage.Ratio..Interest.expense.to.EBIT. Equity.to.Liability
## Min.      :-0.766022                               Min.      :-0.6754
## 1st Qu.   :-0.015100                               1st Qu.   :-0.4619
## Median    :-0.008022                               Median    :-0.2755
## Mean      : 0.006777                               Mean      :-0.0299
## 3rd Qu.   : 0.027756                               3rd Qu.   : 0.1052
## Max.      : 0.576452                               Max.      : 3.7263
```

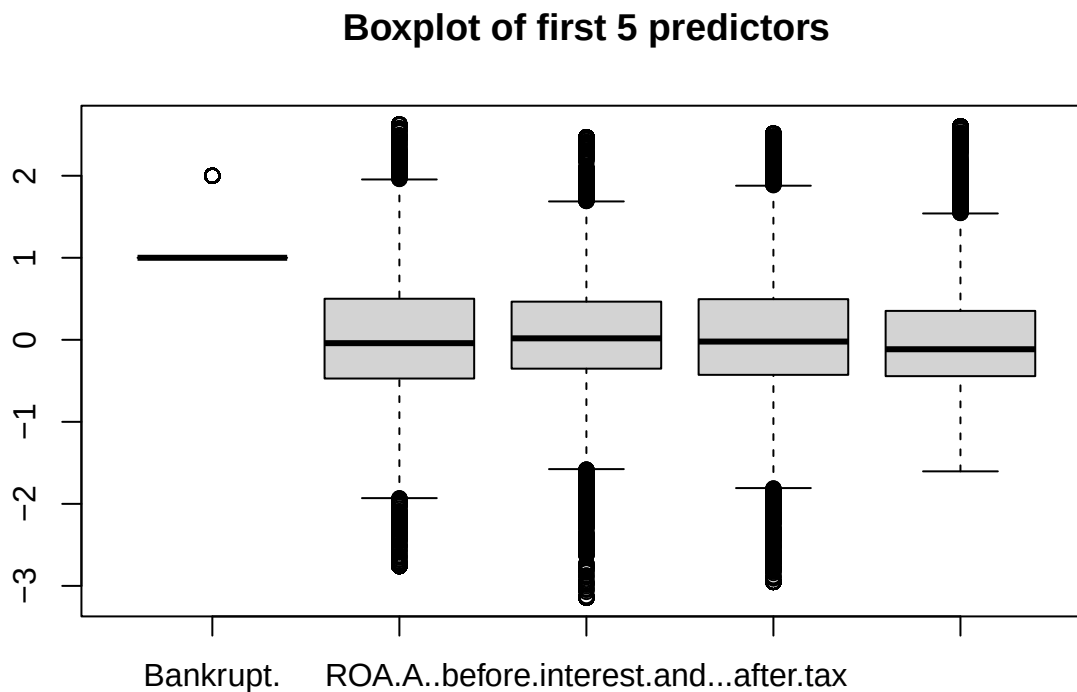
```
num_cols <- sapply(data, is.numeric)
data[num_cols] <- lapply(data[num_cols], as.numeric)
num_cols <- sapply(data, is.numeric)
```

```
cor(data[, num_cols][,1:6])
```

```
## ROA.C..before.interest.and.depreciation.before.interest
## ROA.A..before.interest.and...after.tax
## ROA.B..before.interest.and.depreciation.after.tax
## Operating.Gross.Margin
## Realized.Sales.Gross.Margin
## Operating.Profit.Rate
## ROA.A..before.interest.and...after.tax
## ROA.C..before.interest.and.depreciation.before.interest 0.9433428
## ROA.A..before.interest.and...after.tax 1.0000000
## ROA.B..before.interest.and.depreciation.after.tax 0.9568454
## Operating.Gross.Margin 0.4164613
## Realized.Sales.Gross.Margin 0.4131522
## Operating.Profit.Rate 0.6738588
## ROA.B..before.interest.and.depreciation.after.interest
## ROA.C..before.interest.and.depreciation.before.interest 0.98
## ROA.A..before.interest.and...after.tax 0.95
## ROA.B..before.interest.and.depreciation.after.tax 1.00
## Operating.Gross.Margin 0.43
## Realized.Sales.Gross.Margin 0.42
## Operating.Profit.Rate 0.66
## Operating.Gross.Margin
## ROA.C..before.interest.and.depreciation.before.interest 0.4284938
## ROA.A..before.interest.and...after.tax 0.4164613
## ROA.B..before.interest.and.depreciation.after.tax 0.4305966
## Operating.Gross.Margin 1.0000000
## Realized.Sales.Gross.Margin 0.9990723
## Operating.Profit.Rate 0.3931049
## Realized.Sales.Gross.Margin
## ROA.C..before.interest.and.depreciation.before.interest 0.4255090
## ROA.A..before.interest.and...after.tax 0.4131522
## ROA.B..before.interest.and.depreciation.after.tax 0.4275056
## Operating.Gross.Margin 0.9990723
```

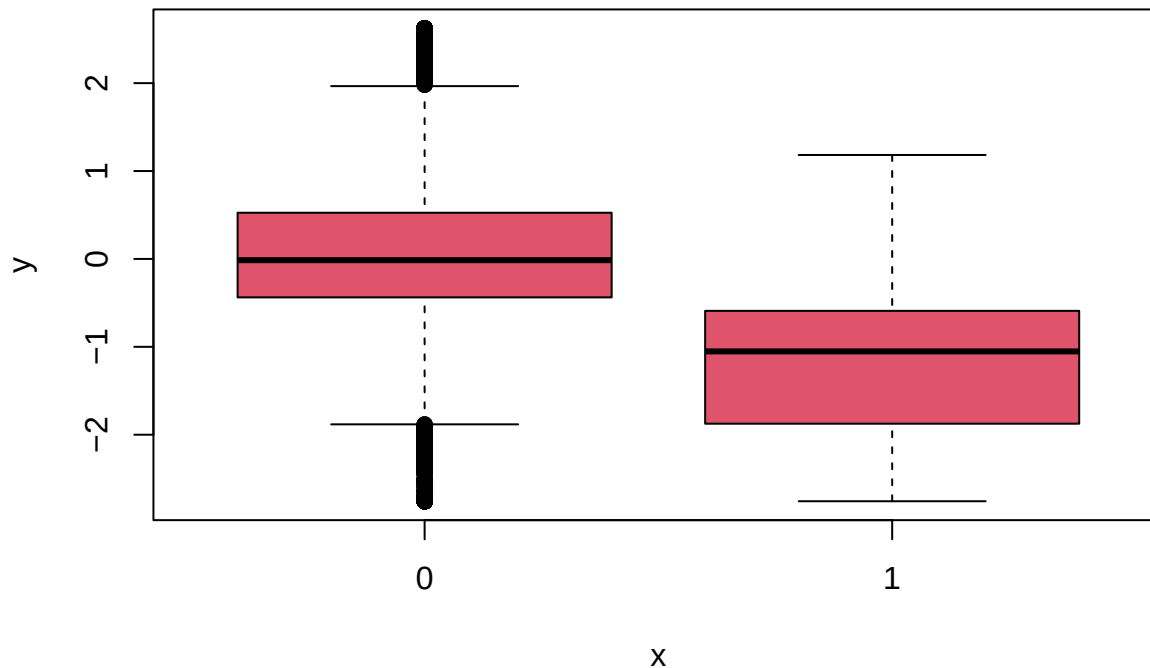
```
## Realized.Sales.Gross.Margin 1.0000000
## Operating.Profit.Rate 0.3903021
## Operating.Profit.Rate
## ROA.C..before.interest.and.depreciation.before.interest 0.6588874
## ROA.A..before.interest.and...after.tax 0.6738588
## ROA.B..before.interest.and.depreciation.after.tax 0.6666277
## Operating.Gross.Margin 0.3931049
## Realized.Sales.Gross.Margin 0.3903021
## Operating.Profit.Rate 1.0000000
```

```
boxplot(data[,1:5],main="Boxplot of first 5 predictors")
```



```
plot(data[,1],data[,2],col=data$Bankrupt.,pch=19,main="Predictor 1 vs Predictor 2 by Bankrupt Status")
```

Predictor 1 vs Predictor 2 by Bankrupt Status



Classification Models

Create Folds

```
set.seed(123)
folds <- createFolds(data$Bankrupt., k=5, list=TRUE, returnTrain=FALSE)
```

Computation

```
metrics <- function(actual, prediction){
  actual <- as.numeric(as.character(actual))
  prediction <- as.numeric(as.character(prediction))
  acc <- mean(actual == prediction)
  tp <- sum(actual == 1 & prediction == 1)
  tn <- sum(actual == 0 & prediction == 0)
  fp <- sum(actual == 0 & prediction == 1)
  fn <- sum(actual == 1 & prediction == 0)
  precision <- ifelse(tp + fp == 0, NA, tp / (tp + fp))
  recall <- ifelse(tp + fn == 0, NA, tp / (tp + fn))
  return(c(Accuracy = acc, Precision = precision, Recall = recall))
}
```

Cross Validation

```
results <- data.frame(Model = character(), Fold = integer(), Accuracy = numeric(),
                      Precision = numeric(), Recall = numeric(), stringsAsFactors=FALSE)

for(i in 1:5){
  test_idx <- folds[[i]]
  train_idx <- setdiff(1:nrow(data), test_idx)

  train <- data[train_idx, ]
  test <- data[test_idx, ]
  y_test <- test$Bankrupt.

  predictors <- setdiff(names(train), "Bankrupt.")
  nzv_train <- nearZeroVar(train[, predictors], saveMetrics=TRUE)
  nzv_cols <- rownames(nzv_train[nzv_train$nzv, ])

  train <- train[, setdiff(names(train), nzv_cols)]
  test <- test[, colnames(train)]

  # KNN
  pred_knn <- knn(train[, setdiff(names(train), "Bankrupt.")],
                  test[, setdiff(names(train), "Bankrupt.")],
                  cl = train$Bankrupt., k = 5)
  res_knn <- metrics(as.numeric(y_test)-1, as.numeric(pred_knn)-1)

  # NaiveBayes
  nb_model <- naiveBayes(Bankrupt.~., data = train)
  pred_nb <- predict(nb_model, test)
  res_nb <- metrics(as.numeric(y_test)-1, as.numeric(pred_nb)-1)

  # GLM
  glm_model <- glm(Bankrupt.~., data = train, family="binomial")
  probs <- predict(glm_model, test, type="response")
  pred_glm <- ifelse(probs > 0.5, 1, 0)
  res_glm <- metrics(as.numeric(y_test)-1, pred_glm)

  temp <- data.frame(
    Model = c("KNN", "NaiveBayes", "GLM"),
    Fold = i,
    Accuracy = c(res_knn["Accuracy"], res_nb["Accuracy"], res_glm["Accuracy"]),
    Precision = c(res_knn["Precision"], res_nb["Precision"], res_glm["Precision"]),
    Recall = c(res_knn["Recall"], res_nb["Recall"], res_glm["Recall"]),
    stringsAsFactors = FALSE
  )

  # Convert numeric columns
  temp$Accuracy <- as.numeric(temp$Accuracy)
  temp$Precision <- as.numeric(temp$Precision)
  temp$Recall <- as.numeric(temp$Recall)

  results <- rbind(results, temp)
}
```

```
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

```
## Warning: glm.fit: algorithm did not converge
```

Evaluate Results

```
final <- aggregate(cbind(Accuracy,Precision,Recall)~Model,data = results, mean)
final
```

```
##      Model Accuracy Precision  Recall
## 1      GLM 0.9677376 0.5026552 0.2727273
## 2      KNN 0.9680309 0.5160957 0.1681818
## 3 NaiveBayes 0.8196031 0.1474879 0.8772727
```